Fiscal Year:	FY 2025	**Task Last Updated:**	FY 06/25/2026
PI Name:	Massa, Gioia Ph.D.		
Project Title:	Pick-and-Eat Salad-Crop Productivity, Nutritional Value, and Acceptability to Supplement the ISS Food System		
Division Name:	Human Research, Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Food and Nutrition**:Risk of Performance Decrement and Crew Illness Due to Inadequate Food and Nutrition		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
PI Email:	gioia.massa@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	321-861-2938
Organization Name:	NASA Kennedy Space Center		
PI Address 1:	ISS Ground Processing and Research		
PI Address 2:	Mail Code UB-A-00		
PI Web Page:			
City:	Kennedy Space Center	**State:**	FL
Zip Code:	32899-0001	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-ILSRA. International Life Sciences Research Announcement
Start Date:	09/01/2015	**End Date:**	09/30/2028
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Douglas, Grace	**Contact Phone:**	
Contact Email:	grace.l.douglas@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 09/30/2028 per T. Sirmons/HHC (Ed., 6/16/26). NOTE: End date changed to 12/31/2025 per PI (Ed., 4/17/26) NOTE: End date changed to 9/30/2025 per HHC element management (Ed., 8/10/21) NOTE: End date changed to 9/30/2021 per PI (Ed., 5/4/2020) NOTE: End date changed to 8/31/2020 per PI (Ed., 8/17/18) NOTE: Element change to Human Health Countermeasures; previously Space Human Factors & Habitability (Ed., 1/18/17) NOTE: Period of performance changed to 9/01/2015-8/31/2018 (previously 7/1/15-6/30/18) per G. Douglas/HRP (Ed., 4/3/16)		
Key Personnel Changes/Previous PI:	To better understand the microbiome and the transcriptome in the VEG-05 experiments we had to add additional collaborators to our team: Dr. Christina Khodadad, Dr. Anirudha Dixit, Cory Spern, and Dr. Natasha Haveman from Kennedy Space Center, and they contributed heavily to this report. Sydney Begerowski and Diana Arias from Johnson Space Center supported the behavioral health components of this study.		

COI Name (Institution):	Douglas, Grace Ph.D. (NASA Johnson Space Center) Hummerick, Mary M.S. (Amentum, Kennedy Space Center) Mitchell, Cary Ph.D. (Purdue University--grant NNX15AN78G) Morrow, Robert Ph.D. (Orbital Technologies Corporation) Wheeler, Raymond Ph.D. (NASA Kennedy Space Center) Young, Millennia Ph.D. (NASA Johnson Space Center) Spencer, LaShelle M.S. (Amentum, Kennedy Space Center) Buncheck, Jess M.S. (Southeastern Universities Research Association, Kennedy Space Center) Bell, Suzanne Ph.D. (NASA Johnson Space Center) Landon, Lauren (Wyle, Johnson Space Center)
Grant/Contract No.:	Internal Project
Performance Goal No.:	
Performance Goal Text:	
Task Description:	The capability to grow nutritious, palatable food for crew consumption during spaceflight has the potential to provide health promoting, bioavailable nutrients, enhance the dietary experience, and reduce launch mass as we move toward longer-duration missions. However, studies of edible produce during spaceflight have been limited, leaving a significant knowledge gap in the methods required to grow safe, acceptable, nutritious crops for consumption in microgravity. The “Veggie” vegetable-production system on the International Space Station (ISS) offers an opportunity to develop a “pick-and-eat” fresh vegetable component to the ISS food system as a first step to bioregenerative supplemental food production. We propose growing salad plants in the Veggie unit during spaceflight, focusing on the impact of light quality and fertilizer formulation on crop morphology, edible biomass yield, microbial food safety, organoleptic acceptability, nutritional value, and behavioral health benefits of the fresh produce. Phase A of the project would involve flight tests using leafy greens. Phase B would focus on dwarf tomato. Our work will help define light colors, levels, and horticultural best practices to achieve high yields of safe, nutritious leafy greens and tomatoes to supplement a space diet of prepackaged food. Our final deliverable will be the development of growth protocols for these crops in a spaceflight vegetable production system. Specific aim 1: Evaluate the effects of four light treatments and two different fertilizer compositions on the yield, morphology, organoleptic acceptability, and nutritional attributes of leafy greens during flight-definition and flight testing. Specific aim 2: Perform cultivar selection and evaluate the effects of four different red: blue light treatments and two different fertilizer compositions on the yield, morphology, organoleptic acceptability, and nutritional attributes of dwarf tomato during ground and flight tests. Specific aim 3: Perform hazard analysis, develop plans for minimizing microbial hazards, and screen flight-grown produce for potential pathogens.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Research Project: Our work on “Pick-and-Eat Salad-Crop Productivity, Nutritional Value, and Acceptability to Supplement the ISS Food System” focuses on developing a fresh food production capability on the International Space Station (ISS). We are using the Veggie hardware to develop light and fertilizer combinations that generate nutritious and appealing leafy green vegetables and dwarf tomatoes that astronauts can safely consume. The results of this research will directly translate to Earth-based controlled environment production of these and similar crops in vertical farms and urban plant factories. The capability to grow nutritious, palatable food for crew consumption during spaceflight can potentially provide health-promoting, bioavailable nutrients, enhance the dietary experience, and reduce launch mass as we move toward longer-duration exploration missions. However, studies of edible produce during spaceflight have been limited, leaving a significant knowledge gap in the methods required to grow safe, acceptable, nutritious crops for consumption in microgravity. The Veggie vegetable-production system on the ISS offers an opportunity to develop a “pick-and-eat” fresh vegetable component to the ISS food system as a first step to bioregenerative supplemental food production. Our goal is to grow salad crops in the Veggie unit during spaceflight and assess the impact of light quality and fertilizer formulation on crop morphology, edible biomass yield, microbial food safety, organoleptic acceptability, nutritional value, and behavioral health benefits. Our work will help define light color ratios, fertilizer composition, and horticultural best practices to achieve high yields of safe, nutritious leafy greens and tomatoes to supplement a space diet of prepackaged food. Our final deliverable will be to develop growth protocols for these crops in a spaceflight vegetable-production system. This will reduce the risk and close the gap of inadequate nutrition by helping us advance bioregenerative food production to supplement the packaged diet for future space exploration.
	JUNE 2026 UPDATE per the PI Pick-and-eat Salad-crop Productivity, Nutritional Value, and Acceptability to Supplement the ISS Food System (VEG-04A, VEG-04B, and VEG-05) is a set of translational experiments with plant research combined with human sensory and behavioral research. The VEG-04 flight experiments conducted on the International Space Station (ISS) in 2019 consisted of growing Mizuna mustard under different lighting spectra, with the first experiment harvesting plants at a single time point and the second using a cut-and-come-again repetitive harvest approach. VEG-05 was conducted Dec. 2022 through March 2023, growing ‘Red Robin’ dwarf tomatoes under the same lighting spectra. In addition to understanding plant growth, development, and produce yields under different lighting conditions, goals for VEG-04/05 were to understand microbial food safety and develop safe handling practices for crops. Human-specific foci included collecting data on behavioral health aspects of crops and crop care and assessing sensory acceptability of consumed crops. Crew were able to consume Mizuna and conduct sensory evaluations for the VEG-04 studies, and space-grown produce acceptability was higher than the same produce consumed in ground-based tests, while light spectrum had less of an impact on acceptability. Due to challenges maintaining adequate moisture for plants during growth of VEG-05 tomatoes, insufficient fruit were formed to allow for crew consumption.

Task Progress:	In VEG-05, samples were taken to conduct plant microbiome and transcriptomic evaluation, as well as planned food safety assessment and elemental analysis. Microbial communities varied between shoot/fruit, and root/adventitious root tissues in tomato. Culture-based pathogen screening, corroborated by 16S rRNA gene and ITS sequencing, yielded negative results. Transcriptomic data for tomato leaves and adventitious roots revealed significant transcriptomic alterations influenced by both spaceflight and lighting spectra. Differential gene expression analysis identified numerous up- and downregulated genes in flight samples compared to ground controls, with adventitious roots displaying pronounced transcriptional changes. Genes associated with stress responses, hormonal signaling, and metabolic pathways were prominently upregulated in spaceflight-grown tissues. Plants grown in spaceflight exhibited extensive adventitious root formation, with gene clusters enriched in antioxidant metabolism, cell wall remodeling, and stress adaptation processes. Leaf tissues demonstrated distinct transcriptional signatures under flight and ground conditions, with blue-rich lighting enhancing stress-responsive pathways and red-rich lighting favoring metabolic stability. Key takeaways from the VEG-04/05 experiments, coupled with other data collected from earlier Veggie crop testing, indicate that crop yield is similar between ground and flight when plants are not stressed. Maintaining plants in a manually watered system such as Veggie is challenging, especially over long durations, and it is hard to avoid water-related stresses. Some differences between ground and flight plant nutrient content have been shown, but these are not consistent between species or experiments and are influenced by experiment timing, crop type, harvest scenario, and other environmental conditions. Microbial counts and compositions vary between flight and ground plants, and microbes typical of the space station environment are generally associated with ISS-grown plants. Human pathogens are rarely identified on space-grown plants and have only been found below levels of concern to human health. The only human pathogens identified in Veggie-grown crops were S. aureus in one plant in the VEG-04B experiment, at very low levels. The safe handling practice approach seems effective at keeping space plant products safe for human consumption. VEG-04/05 data indicate that astronauts enjoy growing and eating plants in spaceflight in general. Astronauts find crop growth tasks generally enjoyable, engaging, meaningful, and stimulating, and they perceive crops as a source of enjoyable sensory stimulation increasingly over time as their missions progress. Overall, plants adapt well to the space environment, and this project demonstrates the potential for space-grown crops. In order to increase the integration of crops in spaceflight, further work is needed on crop selection, hardware, and translation of ground safety and cleaning practices to the resource restrictions of spaceflight. This research was co-funded by the Human Research Program and Space Biology (MTL#1075) in the ILSRA 2015 NRA call. For additional progress on this NASA-sponsored study, please see the Cumulative Bibliography (Ed., 6/22/26).
Bibliography Type:	Description: (Last Updated: 06/24/2026)
Articles in Peer-reviewed Journals	Dixit AR, Khodadad CLM, Schuster JM, Spencer LE, Hummerick ME, Morrow RC, Mitchell CA, Wheeler RM, Massa GD. "Stress and light spectral quality influence the transcriptome of a tomato crop on the International Space Station." BMC Plant Biol. 2026 Apr 7;26(1):764. https://doi.org/10.1186/s12870-025-07621-4 ; PubMed PMID: 41947041; PubMed Central PMCID: PMC13123239 , Apr-2026
Articles in Peer-reviewed Journals	Spern CJ, Hummerick ME, Khodadad CLM, Morales CJ, Dixit AR, Spencer LE, Mitchell C, Morrow B, Wheeler RM, Massa GD. "The microbial communities of a tomato crop grown in Veggie under different lighting regimes on the International Space Station." Life Sci Space Res. 2026 Jan 26. https://doi.org/10.1016/j.lssr.2026.01.010 , Jan-2026
Articles in Peer-reviewed Journals	Landon LB, Begerowski SR, Roma PG, Whiting SE, Bell ST, Massa GD. "Sustaining the Merry Space farmer with pick-and-eat crop production." npj Microgravity. 2025 Oct 27;11(1):72. https://doi.org/10.1038/s41526-025-00513-9 ; PubMed PMID: 41145494; PubMed Central PMCID: PMC12559293 , Oct-2025